Antivirus Software (AV)

LetsDefend SOC Analyst lesson | Asha Latha Panuganti | October 2026

Summary

Hands-on lab triaging Windows Defender antivirus detections: classifying the malware, finding the affected file and judging the real risk from context.

Objectives

- Understand what antivirus logs record: file, threat name, category, severity, action and path.
- Classify detections and decide from context whether they indicate a live infection.

Environment and tools

LetsDefend lab VM (Windows), Notepad++, Windows Defender detection log.

Method

- Read the `Filename`, `Threat Name`, `Category`, `Severity`, `Action` and `Detect Path` fields for each entry.
- Matched each file to its category rather than relying on the threat name alone.
- Used the file path and process name to judge context.

Findings

Log	Key fields observed	Finding
Defender AV	executable.8180.exe: Trojan:Win32/Swrort.A, Category Trojan (8), Severe (5). program1: Backdoor:Linux/GetShell.A!xp, Category Backdoor (6), Severe (5), path Desktop\malware\program1	**executable.8180.exe** is a **Trojan**. The Backdoor detection is the file **program1**.

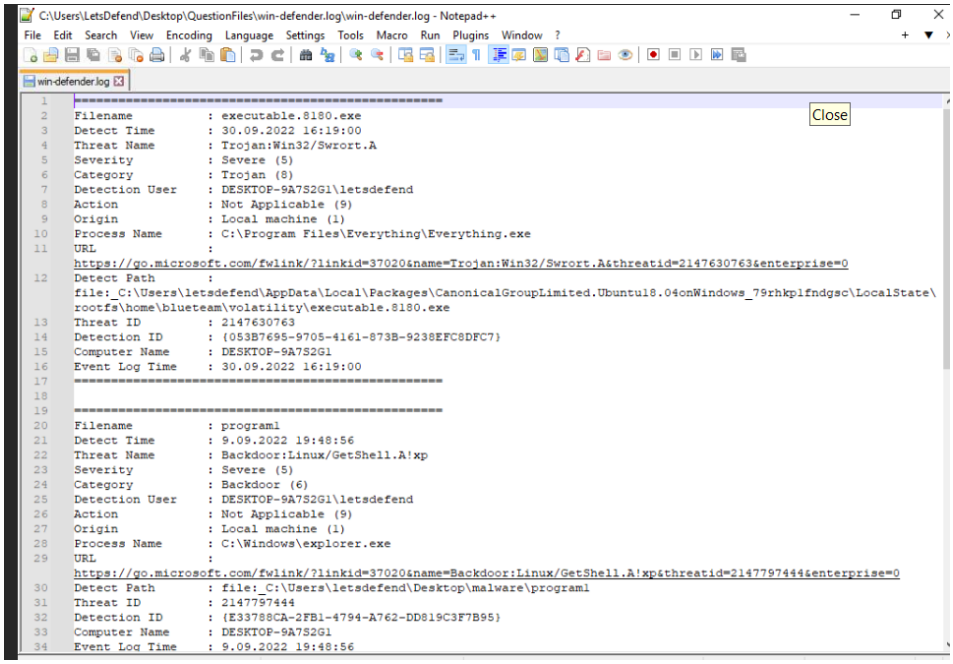


Figure 1: Windows Defender log with Trojan and Backdoor detections.

Security observations

- **Context decides severity.** Defender rated both detections Severe, but the Trojan sat in a Volatility folder inside an Ubuntu (WSL) profile and the backdoor in a folder named malware. Both look like analyst or lab samples, not a live infection.
- **Action Not Applicable.** Both entries show this, so nothing was recorded as removed. An analyst would confirm the files are contained.
- **Cross-platform detections.** A Linux backdoor signature fired on a Windows host, which is expected when Linux binaries are stored or run through WSL.

Skills demonstrated

Antivirus log triage, malware classification, risk judgement from context, documenting findings.

Next steps

- Hash both files and check them against threat intelligence.
- Build a Wazuh dashboard of Defender detections by category and severity.